

Roll Number:

Name:

Section:



Indian Institute of Information Technology, Sri City, Chittoor

Spring Semester Schedule Quiz 1, February-2026

UG-1, Second Sem

Name of the Exam: P & S (S2)

Duration: 20 Minutes

Feb 14, 2026

Max. Marks: 15

Instructions:

1. Calculators are allowed.
 2. All questions are mandatory and carry equal mark.
 3. Tick only the right option. If we find two ticks for one question, '0' marks will be awarded.
 4. Rough work should not be done in the question paper.
 5. Return the question paper along with the rough sheet before leaving the exam hall.
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1. If a random variable X has a Poisson distribution with mean 7, then the expression $E[(X + 2)^2]$ equals

- (a) 88
- (b) 25
- (c) 54
- (d) 45

Answer (a).

2. Which one of the following statements is not true?

- (a) In a negatively skewed distribution, mode $>$ mean $>$ median
- (b) The measure of skewness is dependent upon the amount of dispersion
- (c) In a symmetric distribution, the values of mean, mode and median are the same
- (d) In a positively skewed distribution, mean $>$ median $>$ mode

Answer: (a)

3. The expected number of distinct units in a simple random sample of 3 units drawn with replacement from a population of 100 units is

- (a) $2 + \frac{99^2}{100^3}$

(P.T.O.)

- (b) $3 - \left(\frac{99}{100}\right)^2$
- (c) $100 - \frac{99^3}{100^2}$
- (d) $3 - \left(\frac{99}{100}\right)^3$

Answer: (c).

4. Let E and F be two events such that $0 < P(E) < 1$, $0 < P(F) < 1$, and $P(E \cap F) > P(E)P(F)$. Which of the following statements is TRUE?

- (a) E and F are independent
- (b) $P(F | E^c) > P(F)$
- (c) $P(E | F) > P(E)$
- (d) $P(E | F^c) > P(E)$

Answer: (c)

5. Compute the median for the exponential distribution with parameter λ .

- (a) $\frac{\ln(3)}{\lambda}$
- (b) $\frac{\ln(2)}{\lambda}$
- (c) $\frac{2\ln(2)}{\lambda}$
- (d) $\frac{\ln(2)}{2\lambda}$

Answer: (b)

6. Let the continuous r.v. X denote the weight (in pounds) of a package. The range of weight of packages is between 45 and 60 pounds. Determine the probability that a package weighs more than 50 pounds.

- (a) $\frac{4}{3}$
- (b) $\frac{2}{3}$
- (c) $\frac{1}{3}$
- (d) $\frac{8}{3}$

Answer: (b)

7. The moment generating function of an integer valued random variable X is given by

$$M_X(t) = \frac{1}{10} (2 + e^t + 4e^{2t} + 3e^{3t}) e^{-t}.$$

Then $P(2X + 5 < 7)$ is equal to

- (a) $\frac{4}{10}$
- (b) $\frac{3}{10}$
- (c) 1

(d) $\frac{7}{10}$

Answer: (b)

8. A probability density function on the interval $[a, 1]$ is given by $\frac{1}{x^2}$ and outside this interval the value of the function is zero. The value of a is:

- (a) 0.25
- (b) 0.5
- (c) 0.075
- (d) 0.05

Answer: (b)

9. The probability of getting a “head” in a single toss of a biased coin is 0.3. The coin is tossed repeatedly until a “head” is obtained. If the tosses are independent, then the probability of getting a “head” for the first time on the 5th toss is

- (a) 0.085
- (b) 0.072
- (c) 0.067
- (d) 0.052

Answer: (b)

10. For a random variable X (where $-\infty < x < \infty$) following normal distribution, the mean is $\mu = 100$. If $P(X \geq 110) = \alpha$, then $P(90 \leq X \leq 110)$ is equal to

- (a) $1 - \alpha$
- (b) 2α
- (c) $1 - 2\alpha$
- (d) $1 - \frac{\alpha}{2}$

Answer: (c)

11. Let $\{1, 2, 3, 4\}$ represent equally likely outcomes. Let $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{3, 4\}$, $A_4 = \{1, 2, 3\}$. Which of the following statements is true?

- (a) A_2 and A_4 are independent.
- (b) A_1 and A_4 are not independent.
- (c) A_1 and A_2 are not independent.
- (d) A_3 and A_4 are independent.

Answer: (b)

12. The lifetime of a component is exponentially distributed with parameter 2. The probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is

- (a) 0.43
- (b) 0.37
- (c) 0.27

(d) 0.63

Answer (b)

13. Let X be a random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x+1}{3}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Then $P\left(\frac{1}{3} < X < \frac{3}{4}\right) + P(X = 0)$ **equals**

(a) $\frac{13}{36}$

(b) $\frac{7}{36}$

(c) $\frac{17}{36}$

(d) $\frac{11}{36}$

Answer (c)

14. **Assume that** $X \sim \text{Binomial}(n, p)$ **and** $Y \sim \text{Poisson}(\lambda)$ **with** $E(X) = E(Y)$. **Then**

(a) $\text{Var}(Y) < \text{Var}(X)$

(b) $\text{Var}(X) = \text{Var}(Y)$

(c) $\text{Var}(X)$ may be larger or smaller depending on n, p, λ

(d) $\text{Var}(X) < \text{Var}(Y)$

Answer: (d)

15. **In a population, 2% of people have a genetic condition. A test detects it correctly in 90% of cases and has 5% false positives. If a person tests positive, the probability they actually have the condition is**

(a) $\frac{17}{65}$

(b) $\frac{18}{67}$

(c) $\frac{16}{67}$

(d) $\frac{19}{65}$

Answer: (b)
